Chapter 12

Name _____

CHAPTER 12

Lesson 1

Multiples of Unit Fractions

Book 2 : Page(469)

Time: 1 min

I Can Objective

I can find multiples of a unit fraction by multiplying a unit fraction by a whole number.

Learning Goal

Write a fraction as a product of a whole number and a unit fraction.

Vocabulary

Fraction - a part of a whole

Session 1

02:00

DAILY ROUTINES

Problem of the Day 12.1

Fran makes lasagna and divides it into 12 equal pieces. Her family eats 8 pieces for dinner. There is $\frac{4}{12}$ of the lasagna left. How can you write $\frac{4}{12}$ as the sum of unit fractions?

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03:00

I DO

At a pizza party, each pizza was cut into 6 equal slices. At the end of the party, there was $\frac{5}{6}$ of a pizza left. Roberta put each of the leftover slices in its own freezer bag. How many bags did she use? What part of a pizza did she put in each bag?

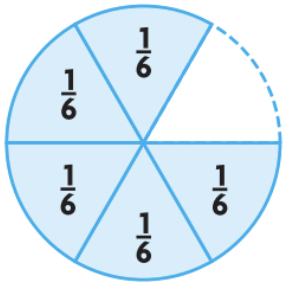
- How many slices of pizza were eaten?

_____

- What fraction of the pizza is 1 slice?

_____

Example **Write $\frac{5}{6}$ as the product of a whole number and a unit fraction.**



The picture shows $\frac{5}{6}$ or

_____ sixth-size pieces.

Each sixth-size pieces of the pizza can be shown by the

unit fraction _____.

You can use unit fractions to show $\frac{5}{6}$ in two ways.

$\frac{5}{6}$ = _____ + _____ + _____ + _____ + _____

$\frac{5}{6}$ = _____ $\times \frac{1}{6}$

The number of addends, or the multiplier, represents the number of bags used.

The unit fractions represent the part of a pizza in each bag.

So, Roberta used _____ bags. She put _____ of a pizza in each bag.

Remember

You can use multiplication to show repeated addition.

3×4 means $4 + 4 + 4$.

4×2 means $2 + 2 + 2 + 2$.

Math Talk

MP **Attend to precision.**

Give an example of how you would write a fraction greater than 1 as a mixed number.

03:00

I DO

UNLOCK the Problem Real World

At a pizza party, each pizza was cut into 6 equal slices. At the end of the party, there was $\frac{5}{6}$ of a pizza left. Roberta put each of the leftover slices in its own freezer bag. How many bags did she use? What part of a pizza did she put in each bag?

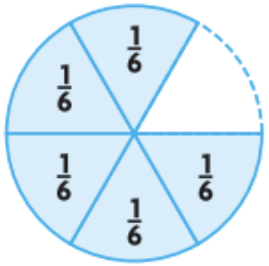
- How many slices of pizza were eaten?

  1

- What fraction of the pizza is 1 slice?

  $\frac{1}{6}$

Example **Write $\frac{5}{6}$ as the product of a whole number and a unit fraction.**



The picture shows $\frac{5}{6}$ or

5 sixth-size pieces.

Each sixth-size pieces of the pizza can be shown by the

unit fraction $\frac{1}{6}$.

Remember

You can use multiplication to show repeated addition.

3×4 means $4 + 4 + 4$.

4×2 means $2 + 2 + 2 + 2$.

You can use unit fractions to show $\frac{5}{6}$ in two ways.

$\frac{5}{6} = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6}$

$\frac{5}{6} = 5 \times \frac{1}{6}$

The number of addends, or the multiplier, represents the number of bags used.

The unit fractions represent the part of a pizza in each bag.

So, Roberta used 5 bags. She put $\frac{1}{6}$ of a pizza in each bag.

Math Talk

MP Attend to precision.

Give an example of how you would write a fraction greater than 1 as a mixed number.

Math Talk: Possible answer: If I wrote $\frac{3}{2}$ as a mixed number, I would divide the numerator by 2 and get 1 with 1 left over. The answer is $1\frac{1}{2}$.

Share and Show

Math Board

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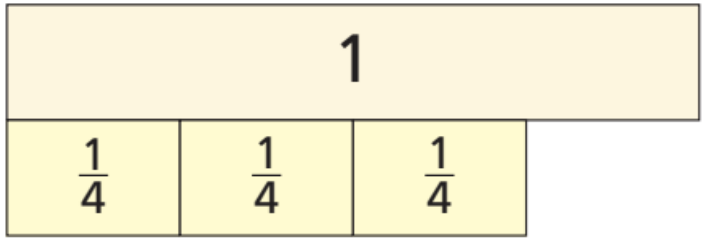
03:00

I DO

1. Use the picture to complete the equations.



$\frac{3}{4} =$ _____ + _____ + _____

$\frac{3}{4} =$ _____ $\times \frac{1}{4}$

Write the fraction as a product of a whole number and a unit fraction.

2. $\frac{4}{5} =$ _____

3. $\frac{3}{10} =$ _____

4. $\frac{8}{3} =$ _____

List the next four multiples of the unit fraction.

5. $\frac{1}{6}$, ____, ____, ____, ____

6. $\frac{1}{3}$, ____, ____, ____, ____

Share and Show

Math Board

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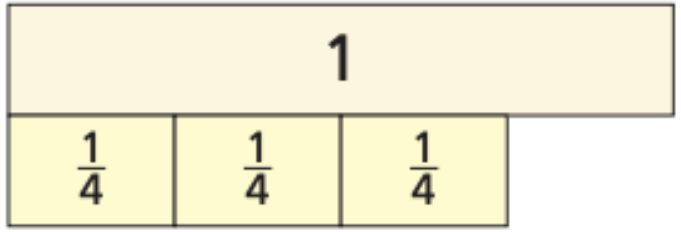
03:00

I DO

1. Use the picture to complete the equations.



$\frac{3}{4} = \frac{1}{4} + \frac{1}{4} + \frac{1}{4}$

$\frac{3}{4} = 3 \times \frac{1}{4}$

Write the fraction as a product of a whole number and a unit fraction.

2. $\frac{4}{5} = 4 \times \frac{1}{5}$

3. $\frac{3}{10} = 3 \times \frac{1}{10}$

4. $\frac{8}{3} = 8 \times \frac{1}{3}$

List the next four multiples of the unit fraction.

5. $\frac{1}{6}$, $\frac{2}{6}$, $\frac{3}{6}$, $\frac{4}{6}$, $\frac{5}{6}$

6. $\frac{1}{3}$, $\frac{2}{3}$, $\frac{3}{3}$, $\frac{4}{3}$, $\frac{5}{3}$

Math Talk: Possible explanation: $\frac{8}{5}$ can be written as the product of a counting number and $\frac{1}{5}$, $8 \times \frac{1}{5} = \frac{8}{5}$.

Math Talk

MP Reason abstractly and

Session 1

Problem Solving · Applications

Real World

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03:00

I DO

12. (MP) Ailee uses $\frac{1}{2}$ cup of blueberries to make each loaf of blueberry bread. Explain how many loaves of blueberry bread she can make with $2\frac{1}{2}$ cups of blueberries.

13. Sinna cut a loaf of bread into 12 equal slices. His family ate some of the bread and now $\frac{5}{12}$ of the loaf is left. Sinna wants to put each of the leftover slices in its own bag. How many bags does Sinna need?

14. Which fraction is a multiple of $\frac{1}{5}$? Mark all that apply.

- ○ $\frac{4}{5}$
- ○ $\frac{5}{12}$
- ○ $\frac{5}{10}$
- ○ $\frac{3}{5}$

Problem Solving · Applications

Real World

Book 2 : Page(469)

03:00

I DO

12. (MP) Ailee uses $\frac{1}{2}$ cup of blueberries to make each loaf of blueberry bread. Explain how many loaves of blueberry bread she can make with $2\frac{1}{2}$ cups of blueberries.

5 loaves; Possible explanation: $2\frac{1}{2}$ can be written as $\frac{5}{2}$, which is the same as $5 \times \frac{1}{2}$.

13. Sinna cut a loaf of bread into 12 equal slices. His family ate some of the bread and now $\frac{5}{12}$ of the loaf is left. Sinna wants to put each of the leftover slices in its own bag. How many bags does Sinna need?

5 bags

14. Which fraction is a multiple of $\frac{1}{5}$? Mark all that apply.

- ● $\frac{4}{5}$
- ○ $\frac{5}{12}$
- ○ $\frac{5}{10}$
- ● $\frac{3}{5}$

Practice and Homework

Book 2 : Page(723)

Goal: Apply division to word problems + reasoning

Multiples of Unit Fractions

Practice and Homework

Book 2 : Page(473)

03:00

I DO

Write the fraction as a product of a whole number and a unit fraction.

1. $\frac{5}{6} =$ $5 \times \frac{1}{6}$

2. $\frac{7}{8} =$ _____

3. $\frac{5}{3} =$ _____

4. $\frac{9}{10} =$ _____

5. $\frac{3}{4} =$ _____

6. $\frac{11}{12} =$ _____

List the next four multiples of the unit fraction.

7. $\frac{1}{5}$, ____, ____, ____, ____

8. $\frac{1}{8}$, ____, ____, ____, ____

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03:00

I DO

Write the fraction as a product of a whole number and a unit fraction.

1. $\frac{5}{6} =$ $5 \times \frac{1}{6}$

2. $\frac{7}{8} =$ $7 \times \frac{1}{8}$

3. $\frac{5}{3} =$ $5 \times \frac{1}{3}$

4. $\frac{9}{10} =$ $9 \times \frac{1}{10}$

5. $\frac{3}{4} =$ $3 \times \frac{1}{4}$

6. $\frac{11}{12} =$ $11 \times \frac{1}{12}$

List the next four multiples of the unit fraction.

7. $\frac{1}{5}$, $\frac{2}{5}$, $\frac{3}{5}$, $\frac{4}{5}$, $\frac{5}{5}$

8. $\frac{1}{8}$, $\frac{2}{8}$, $\frac{3}{8}$, $\frac{4}{8}$, $\frac{5}{8}$

Problem Solving Real World

Book 2 : Page(473)

03:00

I DO

9. So far, Midori has read $\frac{5}{6}$ of a book. She has read the same number of pages each day for 5 days. What fraction of the book does Midori read each day?

_____

10. Nicholas buys $\frac{3}{8}$ pound of cheese. He puts the same amount of cheese on 3 sandwiches. How much cheese does Nicholas put on each sandwich?

_____

Problem Solving Real World

Book 2 : Page(473)

03:00

I DO

9. So far, Midori has read $\frac{5}{6}$ of a book. She has read the same number of pages each day for 5 days. What fraction of the book does Midori read each day?

$\frac{1}{6}$ of the book

10. Nicholas buys $\frac{3}{8}$ pound of cheese. He puts the same amount of cheese on 3 sandwiches. How much cheese does Nicholas put on each sandwich?

$\frac{1}{8}$ pound

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03:00

I DO

Lesson Check

12. Selena walks from home to school each morning and back home each afternoon. Altogether, she walks $\frac{2}{3}$ mile each day. How far does Selena live from school?

_____

13. Will uses $\frac{3}{4}$ cup of olive oil to make 3 batches of salad dressing. How much oil does Will use for one batch of salad dressing?

_____

Spiral Review

14. Liza bought $\frac{5}{8}$ pound of trail mix. She gives $\frac{2}{8}$ pound of trail mix to Michael. How much trail mix does Liza have left?

15. Ximena has a piece of rope that is $6\frac{2}{3}$ feet long. How do you write $6\frac{2}{3}$ as a fraction greater than 1?

Book 2 : Page(474)

03:00

I DO

Lesson Check

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$\frac{1}{3}$ mile

13. Will uses $\frac{3}{4}$ cup of olive oil to make 3 batches of salad dressing. How much oil does Will use for one batch of salad dressing?

$\frac{1}{4}$ cup

Spiral Review

14. Liza bought $\frac{5}{8}$ pound of trail mix. She gives $\frac{2}{8}$ pound of trail mix to Michael. How much trail mix does Liza have left?

$\frac{3}{8}$ pound

15. Ximena has a piece of rope that is $6\frac{2}{3}$ feet long. How do you write $6\frac{2}{3}$ as a fraction greater than 1?

$\frac{20}{3}$

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03:00

I DO

16. A group of students makes a pattern with their house numbers. Kyrie's house number is missing. What is Kyries house number?

29, 39, ?, 59, 69, 79

17. Blima buys 12 cupcakes. Nine of the cupcakes have chocolate frosting and the rest have vanilla frosting. What fraction of the cupcakes have vanilla frosting?

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03:00

I DO

16. A group of students makes a pattern with their house numbers. Kyrie's house number is missing. What is Kyries house number?

29, 39, ?, 59, 69, 79

49

17. Blima buys 12 cupcakes. Nine of the cupcakes have chocolate frosting and the rest have vanilla frosting. What fraction of the cupcakes have vanilla frosting?

$\frac{3}{12}$ or $\frac{1}{4}$